saamehsanaaee.github.io
0000-0002-8858-9117
saameh-sanaaee
saamehsanaaee

# Saameh Sanaaee

## Education

Oct 2026 – Present **M.Sc. Cellular and Molecular Biology**, ***University of Tehran***, College of Interdisciplinary Science and Technology (CIST), Tehran, Iran

Sep 2019 – Feb 2024 **B.Sc. Cellular and Molecular Biology**, ***Islamic Azad University of Medical Sciences***, Tehran, Iran
GPA: 16.46/20.00

## Experience

Oct 2024 – May 2025 **Scholar in the Impact Scholars Program**, ***Neuromatch Academy***, Neuromatch Inc., US (Remote), Project Advisor: Raul Rodriguez Cruces

- Developed statistical and deep learning models, trained on and tested with task-fMRI data.
- Designed and created models that elucidate architecture of working memory and act as sensors for task demand during non-working-memory task engagement.
- Member of a team of Impact Scholars selected by Neuromatch Academy to continue preliminary projects after the Computational Neuroscience 2024 Summer Course.

Apr 2023 – Jul 2024 **Researcher**, ***NeuroTRACT***, Tehran, Iran

- Collaborated as a researcher in a neurobiological research published in the Archives of Gerontology and Geriatrics.
- Was responsible for literature review, writing, editing, and proofreading.

Sep 2019 – Dec 2019 **Laboratory Assistant**, ***Gen Pajuhan Ebne Sina Laboratory***, Tehran, Iran

- Conducted plasmid isolation, PCR, and gel electrophoresis in the molecular lab.

## Publications

2025 **Sanaaee, S.**, Mu, B., Tang, C. K.-M., Caslick-Waller, Z. R. G., Li, W., Gao, T., & Rodriguez Cruces, R. (2025). *Parallel GNN-LSTM model predicting working memory involvement during language and emotion processing.* `https://doi.org/10.5281/ZENODO.15126505`

2024 Farhadieh, M.-E., Mozafar, M., **Sanaaee, S.**, Sodeifi, P., Kousha, K., Zare, Y., Zare, S., Maleki Rad, N., Jamshidi-Goharrizi, F., Allahverdloo, M., Rahimmi, A., Sadeghi, M., Shafie, M., & Mayeli, M. (2024). Polygenic hazard score predicts synaptic and axonal degeneration and cognitive decline in Alzheimer's disease continuum. *Archives of Gerontology and Geriatrics, 127*, 105576. `https://doi.org/10.1016/j.archger.2024.105576`

## Presentations

2025 **Working Memory Involvement in Higher Cognition: Insights from fMRI Modeling**, *Impact Scholars Program Seminar Presentations*, Neuromatch Academy, Oral Presentation
`https://youtu.be/F5_z5fJTgNs`

## Projects

Jul 2024 – Present **Working Memory Architecture and Demand (WoMAD) Model**
Developing neural networks analyzing task-fMRI and BOLD timeseries data to examine the structure of human working memory and its involvement in non-working memory tasks.
`https://github.com/saamehsanaaee/WoMAD`

## Skills

**Wet Lab**

**Molecular Laboratory Techniques**

- DNA Extraction, RNA and Plasmid Isolation, Gel Electrophoresis, Western Blotting
- Polymerase Chain Reaction (PCR), Real-Time Polymerase Chain Reaction (qPCR, RT-qPCR)
- Enzyme-Linked Immunosorbent Assay (ELISA), BCA Protein Assay, Affinity Chromatography
- MTT assay, Direct and indirect MTT assay for tissue engineering scaffolds

**Cell and Animal Laboratory Techniques**

- Mammalian Cell Culture, Cancer and Stem Cell Culture, 3D Cell Culture, Bacterial Cell Culture
- *in vivo* Tissue Grafting, Primary Stem Cell Isolation from Laboratory Rodents
- Clean, Aseptic, and Sterile Procedures

**Histology and Microscopy**

- Multi-channel Immunofluorescence (IF) Microscopy, H&E and Special Tissue/Cell Staining, Immunohistochemistry (IHC)
- Karyotyping, Tissue processing, Histopathological Techniques

**Dry Lab**

**Computational Techniques and Tools**

- Primer Design for PCR and qPCR, Next-Generation Sequencing (NGS) Analysis
- Machine Learning, Deep Learning, Neural Networks, and Computational Modeling for Medical Image/Signal Analysis
- **Software:** GeneRunner, Gene5™, StepOne™, ImageJ, Connectome Workbench, CAD

**Programming**

- **Python** (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Nibabel, Nilearn)
- **Bash, Git, R**, HTML, CSS

## Certifications and Training

2025 **Time Series Analysis**, *Hasso Plattner Institute (HPI)*, openHPI MOOC, *Taught by Mario Tormo Romero*
`https://open.hpi.de/verify/xibad-pamag-tovel-medak-vafab`

2024 **SciComms Workshop**, *Neuromatch Acadmey*, Virtual Workshop, *Taught by Xaq Pitkow*

2024 **Computational Neuroscience**, *Neuromatch Academy*, Interactive Course and Project
`https://portal.neuromatchacademy.org/certificate/62c18dc2-0fb4-4dce-a403-31eab01b0a0d`

2023 **Tissue Engineering & Regenerative Medicine**, *GenIran Research & Education Laboratory*, In-person Laboratory Training
Grade: 100%

2023 **Cell Modeling in Systems Biology**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Genetic Engineering and Molecular Cloning**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Applied Microbiology**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Immunoassay Techniques; Principles & Troubleshooting**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Cytogenetics & Karyotyping techniques**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Histopathology Techniques**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2022 **Human/Cancer Cell Culture Technique and MTT Assay**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2021 **Introduction to Bioinformatics**, *GenIran Research & Education Laboratory*, Online Training

2021 **Essential Lab Skills and Molecular Genetic Techniques**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2021 **Real-Time PCR**, *GenIran Research & Education Laboratory*, In-person Laboratory Training

2021 **SDS-PAGE and Western Blot**, *GenIran Research & Education Laboratory*, In-person Laboratory Training